CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two trains cover the same distance at speeds in the ratio 5 : 9. The ratio of the times they take is: 1

(a) 25 : 81 **(b)** 5 : 9 **(c)** 14 : 5 **(d)** 9 : 5

2. If 5 : 6 = 25 : 30, then (5+6) : 6 = x : 30, where x is: 1

(a) 25 **(b)** 30 **(c)** 61 **(d)** 55

3. The sides of two squares are in the ratio 4 : 7. The ratio of their areas is: 1

(a) 16 : 7 **(b)** 4 : 7 **(c)** 4 : 49 **(d)** 16 : 49

4. The number x for which 5 : 8 = 15 : x is: 1

(a) 24 **(b)** 28 **(c)** 19 **(d)** 23

5. If x : y = 2 : 5 and y : z = 10 : 3, then x : y : z is: 1

(a) 20 : 50 : 15 **(b)** 4 : 10 : 3 **(c)** 2 : 15 : 3 **(d)** 2 : 5 : 3

6. The present ages of two brothers are in the ratio 7 : 5. After 3 years the ratio becomes 4 : 3. The present age of the elder brother is: 1

(a) 36 years **(b)** 15 years **(c)** 24 years **(d)** 21 years

7. ₹ 1880 is divided in parts proportional to $\frac{1}{2}$, $\frac{1}{5}$ and $\frac{1}{10}$. The largest part is: 1

(a) ₹ 1175 **(b)** ₹ 235 **(c)** ₹ 376 **(d)** ₹ 470

8. 8 men can finish a job in 30 days. After 10 days, 2 more men join. The remaining work will be finished in: 1

(a) 18 days **(b)** 16 days **(c)** 24 days **(d)** 20 days

9. **Assertion (A):** If ₹1,200 is divided between A and B in the ratio $3 : 5$, B gets ₹750. 1

Reason (R): B's share is $\frac{5}{3+5}$ of the total amount.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The ratio of 40 minutes to 2 hours is $1 : 3$. 1
Reason (R): The ratio of two quantities of the same kind has no unit.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Divide ₹ 1500 into parts proportional to $\frac{1}{2}$ and $\frac{1}{3}$. 2

12. The sides of two squares are in the ratio 2 : 9. Find the ratio of their perimeters and of their areas. 2

13. If x : y = 13 : 7, find the ratio (x + y) : (x - y). 2

14. Divide ₹ 1800 into parts proportional to $\frac{1}{4}$ and $\frac{1}{5}$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The ratio of the present ages of a father and son is 5 : 3. After 6 years the ratio will be 3 : 2. Find their present ages. 3

16. 20 men can finish a job in 15 days. After 5 days, 5 more men join them. In how many more days is the job finished? 3

17. Divide ₹ 2400 into three parts proportional to $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{6}$. 3

OR

The ratio of the present ages of a father and son is 8 : 3. After 4 years the ratio will be 9 : 4. Find their present ages. 3

18. 12 men can finish a job in 24 days. After 6 days, 6 more men join them. In how many more days is the job finished? 3

OR

Divide ₹ 2850 into three parts proportional to $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{5}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A prize of ₹15,000 won by a team is shared among three friends A, B and C in the ratio $2 : 3 : 5$. 5
(i) Find the total number of parts.
(ii) Find A's share.
(iii) Find C's share.
(iv) Find B's share, and also B's share if the prize had been ₹24,000 shared in the same ratio.

20. A, B and C invest ₹ 3000, ₹ 5000 and ₹ 2000 for 8, 4 and 6 months respectively. A profit of ₹ 1540 is shared in the ratio of (investment $\times$ time). Find the ratio and each share. 5

OR

A vessel holds 120 L of a mixture of milk and water in the ratio 5 : 3. 24 L of the mixture is removed and replaced by 24 L of water. Find the new ratio of milk to water. 5

END OF QUESTION PAPER — Total: 40 Marks